

6(a)	<table border="1"> <thead> <tr> <th data-bbox="336 215 817 279">feature</th><th data-bbox="817 215 1131 279">DNA</th><th data-bbox="1131 215 1937 279">collagen</th></tr> </thead> <tbody> <tr> <td data-bbox="336 279 817 343">elements</td><td data-bbox="817 279 1131 343">C H O N P ;</td><td data-bbox="1131 279 1937 343">CHON</td></tr> <tr> <td data-bbox="336 343 817 438">monomers</td><td data-bbox="817 343 1131 438">(DNA) nucleotide(s) ;</td><td data-bbox="1131 343 1937 438">amino acid(s) ; ignore any named amino acid</td></tr> <tr> <td data-bbox="336 438 817 510">bonds between monomers</td><td data-bbox="817 438 1131 510">phosphodiester ;</td><td data-bbox="1131 438 1937 510">peptide ;</td></tr> <tr> <td data-bbox="336 510 817 654">site of production in eukaryotic cells</td><td data-bbox="817 510 1131 654">nucleus</td><td data-bbox="1131 510 1937 654">ribosomes / rough endoplasmic reticulum / RER / rough ER ; A cytoplasm / Golgi (body / apparatus / complex)</td></tr> </tbody> </table>	feature	DNA	collagen	elements	C H O N P ;	CHON	monomers	(DNA) nucleotide(s) ;	amino acid(s) ; ignore any named amino acid	bonds between monomers	phosphodiester ;	peptide ;	site of production in eukaryotic cells	nucleus	ribosomes / rough endoplasmic reticulum / RER / rough ER ; A cytoplasm / Golgi (body / apparatus / complex)	6
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6(b)	<i>any two from:</i> 1 two membranes forming, nuclear envelope ; A nuclear envelope is composed of a double membrane 2 nuclear pores too small to allow DNA to, pass through / escape ; A ora 3 AVP ; e.g. prevents DNA from being broken down (by enzymes) I DNA damage by enzymes protects DNA from hydrolytic enzymes <i>idea of</i> nuclear envelope is a barrier, between / that separates, nucleus and, cytoplasm / rest of cell	2															

Question	Answer	Marks
6(c)	any two from: either 1 allow alveoli, to stretch <u>and</u> recoil (during inhalation and exhalation) or allow, lung (tissue) / named airway, to expand and recoil ; I 'contract', 'relax' and 'gas exchange' / 'respiration' 2 prevent <u>alveoli</u>, over-stretching / bursting ; 3 <u>recoils</u> to move air out of <u>alveoli</u> ; A force / expel / push / AW A alveolar duct	2
6(d)	interphase / S (phase) / synthesis (phase) ; A S1 / S2 R if 'interphase + G1 / G2' or 'S + G1 / G2'	1